Worked practice

These five problems ask you to construct calculations using Lectures 7, 8, 11, 12 and 13. Questions come first, followed by hints, short answers and full solutions. Write an attempt before opening the feedback. The course's conventions apply: the top wire is the first tensor factor and most significant bit; phase estimation uses the inverse of the positive-exponent QFT; a Pauli measurement bit b means eigenvalue $(-1)^b$.

Questions

1. A degenerate eigenphase

Use the exact phase-estimation circuit of Theorem 7.4 with three counting qubits, initially in $|000\rangle$, and a two-qubit target. The target's computational basis is ordered 00, 01, 10, 11. The operator and target input are

$$U = \begin{pmatrix} \mathrm{i} & 0 & 0 & 0 \\ 0 & \mathrm{i} & 0 & 0 \\ 0 & 0 & -\mathrm{i} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

$$\psi = \frac{|00\rangle + 2\mathrm{i}\,|01\rangle + \sqrt{3}\,|10\rangle}{\sqrt{8}}.$$

The counting wires control U^4, U^2, U, from top to bottom. The inverse QFT begins with its wire-reversal swaps.

1. Write the state just before the inverse QFT, using an integer $x \in \{0, \ldots, 7\}$ for the counting register.
2. Find the exact joint state after the inverse QFT. List every possible counting outcome, its probability and the conditional target state.
3. Explain why adding the probabilities of the two target basis states with eigenphase $1/4$ gives the outcome weight, but does not replace their conditional superposition by a mixture.

2. Process an order-finding record

An order-finding run from Lecture 8 uses the following data:

$$N = 35,$$
$$a = 2,$$
$$t = 13,$$
$$y = 683.$$

The target starts in $|1\rangle$. Neither the order r nor a latent phase numerator s is supplied.

1. Use Euclid's algorithm to find the continued fraction of $y/2^t$. Compute all its convergents, in order.
2. Test every convergent denominator $q < N$ by exponentiation modulo N. Check that the first successful denominator is the order, rather than merely a multiple of it.

3. Complete the parity, half-power and gcd tests. State whether this record supplies proper factors of N.
4. After the modular tests, compare the recovered fraction with the recorded ratio. Explain what the reading does and does not certify about an unobserved eigenphase.

3. Decode a Steane syndrome

Use these ordered generators of Steane's code from Lecture 12, with qubit 1 on the left:

$$g_1 = IIIXXXX,$$
$$g_2 = IXXIIXX,$$
$$g_3 = XIXIXIX,$$
$$g_4 = IIIZZZZ,$$
$$g_5 = IZZIIZZ,$$
$$g_6 = ZIZIZIZ.$$

The input is an arbitrary code vector ψ. At most one qubit suffers a Pauli error $E \in \{\mathbb{1}, X_j, Y_j, Z_j : 1 \le j \le 7\}$. Ideal generator measurements give the ordered syndrome 110110, where a bit 1 means anticommutation with the error.

1. Derive the syndrome for each type X_j, Z_j and Y_j from the generator strings. Use their three-bit columns to identify E.
2. Specify a correction and show that measurement followed by correction returns ψ for every code vector, not only for a logical basis state.

4. Build a complete recovery channel

The repetition code R is

$$C = \mathrm{span}\{|000\rangle, |111\rangle\}.$$

Use the error list $E_0 = \mathbb{1}$, $E_1 = X_2$ and the construction in Theorem 11.2. Let Π_C be the code projector.

1. Compute the Knill–Laflamme matrix. Find the two error-subspace projectors and their orthogonal complement $\Pi_\perp$.
2. Write Kraus operators for a recovery on the entire eight-dimensional register. Prove completeness, including the complement branch.
3. Prove that this one recovery corrects every channel with Kraus operators $K_k = a_k\mathbb{1} + b_k X_2$, assuming $\sum_k K_k^\dagger K_k = \mathbb{1}$. Preserve an arbitrary logical density matrix.
4. Apply the recovery to the coherent unitary error $V = (\sqrt{3}/2)\mathbb{1} - (\mathrm{i}/2)X_2$ on $\psi = \alpha|000\rangle + \beta|111\rangle$, where $|\alpha|^2 + |\beta|^2 = 1$. Give the syndrome probabilities and the corrected states. Exhibit a nonzero vector in the complement branch.

5. Update signed generators

A three-qubit stabilizer state is specified by the independent commuting generators

$$g_1 = -XXX,$$
$$g_2 = ZZI,$$
$$g_3 = IZZ.$$

Measure $g = IYI$, recording bit b for eigenvalue $(-1)^b$. Use the update rule from Lecture 13 in Lemma 13.3 with pivot g_1.

1. Reconstruct a normalised initial state from the generators.
2. Identify every anticommuting row. Multiply the necessary rows by the pivot, keeping all Pauli phases, and give the updated generators for both values of b.
3. Compute both conditional state vectors and outcome probabilities. Check the updated generators directly on those states.
4. Next measure $h = -YIX$. Find its outcome bit c from the updated generators, including its dependence on b.

Hints

1

Group the target amplitudes by eigenvalue before applying Theorem 7.6. A counting measurement projects onto an entire eigenspace. Its weight is the squared norm of that projection.

2

Start by dividing 8192 by 683. In the convergent recurrence use $p_{-2} = 0$, $p_{-1} = 1$, $q_{-2} = 1$, $q_{-1} = 0$. A denominator is a candidate, not an answer, until its modular power is tested. To verify minimality of a successful denominator, test its proper divisors.

3

The first three generators are X-type, the last three Z-type. A one-qubit Y anticommutes with either nonidentity letter. Read each three-bit generator column as a binary label for its physical qubit.

4

List the basis states in C and X_2C. Their orthogonality makes the KL matrix diagonal. The adjoints of the isometries return each error copy to C. Include the projector onto all remaining basis states as a Kraus operator, and test each recovery operator on both errors before expanding an arbitrary K_k.

5

Use $ZX = \mathrm{i}Y$, $YX = -\mathrm{i}Z$ and $XY = \mathrm{i}Z$. Multiply the old rows by the old pivot before replacing it by the signed measured operator. For the follow-up measurement, multiply two updated rows and compare the full signed operator with h.

Short answers

1

The outcomes are 010 with probability $5/8$ and target $(|00\rangle + 2\mathrm{i}\,|01\rangle)/\sqrt{5}$, and 110 with probability $3/8$ and target $|10\rangle$.

2

The continued fraction is $[0; 11, 1, 169, 1, 3]$. Tests with $q = 1, 11, 12$ give residues $2, 18, 1$. The order is 12, its half-power is 29 modulo 35, and the gcds give 7 and 5.

3

Both nonzero three-bit halves are 110, identifying Y_6. The correction is Y_6.

4

The KL matrix is $\mathbb{1}_2$. Recovery Kraus operators are Π_C, $\Pi_C X_2$ and $\Pi_\perp = \mathbb{1} - \Pi_C - X_2\Pi_C X_2$. The coherent-error branches have probabilities $3/4$ and $1/4$ and both return the original logical state. The vector $|001\rangle$ lies in the complement.

5

The initial state is $(|000\rangle - |111\rangle)/\sqrt{2}$. The updated list is $(-1)^b IYI$, YYX, XYY. Each first outcome has probability $1/2$. The follow-up result is $c = b \oplus 1$ with certainty.

Full solutions

1. Project onto the eigenspace

A binary string $x_1x_2x_3$ represents $x = 4x_1 + 2x_2 + x_3$. After the counting Hadamards and controlled powers, the joint state is

$$\Psi_{\text{before}} = \frac{1}{\sqrt{8}} \sum_{x=0}^{7} |x\rangle \otimes U^x \psi.$$

The target evolution separates into two eigenphases:

$$U^x \psi = \frac{\mathrm{i}^x (|00\rangle + 2\mathrm{i}\,|01\rangle) + \sqrt{3}(-\mathrm{i})^x\,|10\rangle}{\sqrt{8}}.$$

The eigenvalues i and $-\mathrm{i}$ have phases $1/4$ and $3/4$. Both are exact three-bit phases. The inverse QFT therefore sends their counting states to $|010\rangle$ and $|110\rangle$, respectively. The exact output is

$$\begin{aligned}\Psi_{\text{out}} = \frac{1}{\sqrt{8}}\,|010\rangle \otimes (|00\rangle + 2\mathrm{i}\,|01\rangle) \\ + \sqrt{\frac{3}{8}}\,|110\rangle \otimes |10\rangle\,.\end{aligned}$$

The two target components in the first term are orthogonal. Its squared norm is $(1+4)/8 = 5/8$; the second term has squared norm $3/8$. Dividing by the corresponding square roots gives

$$\begin{aligned}\psi_{010} = \frac{|00\rangle + 2\mathrm{i}\,|01\rangle}{\sqrt{5}}. \\ \psi_{110} = |10\rangle\,.\end{aligned}$$

No other counting outcome is possible. Measurement distinguishes the eigenphases, not the two basis vectors inside the same eigenspace. The relative coefficient 2i survives in ψ_{010}. In projector notation the first probability is $\left\|\Pi_{1/4}\psi\right\|^2 = 5/8$; it is the weight of the full eigenspace.

2. Verify each candidate

The counting denominator is $2^{13} = 8192$. Euclid's divisions are

$$\begin{aligned}8192 &= 11 \cdot 683 + 679, \\ 683 &= 1 \cdot 679 + 4, \\ 679 &= 169 \cdot 4 + 3, \\ 4 &= 1 \cdot 3 + 1, \\ 3 &= 3 \cdot 1 + 0.\end{aligned}$$

Since the ratio is less than one, its leading coefficient is zero. Thus

$$\frac{683}{8192} = [0; 11, 1, 169, 1, 3].$$

For each coefficient c_j, the recurrence is

$$p_j = c_j p_{j-1} + p_{j-2},$$
$$q_j = c_j q_{j-1} + q_{j-2}.$$

The initial values are

$$p_{-2} = 0,$$
$$p_{-1} = 1,$$
$$q_{-2} = 1,$$
$$q_{-1} = 0.$$

These recurrences give every convergent:

j	c_j	p_j	q_j
0	0	0	1
1	11	1	11
2	1	1	12
3	169	170	2039
4	1	171	2051
5	3	683	8192

For example, the fourth row uses $170 = 169 \cdot 1 + 1$ and $2039 = 169 \cdot 12 + 11$. Only the first three denominators are less than 35. Their modular tests are

$$2^1 \equiv 2 \pmod{35},$$
$$2^{11} \equiv 18 \pmod{35},$$
$$2^{12} \equiv 1 \pmod{35}.$$

The repeated squares are $2^2 \equiv 4$, $2^4 \equiv 16$ and $2^8 \equiv 11$ modulo 35. In particular, $2^{11} \equiv 11 \cdot 4 \cdot 2 \equiv 18$ and $2^{12} \equiv 11 \cdot 16 \equiv 1$ modulo 35. The order divides the successful exponent 12. Its proper positive divisors are $1, 2, 3, 4, 6$, whose residues are $2, 4, 8, 16, 29$. None is one, so the order is $r = 12$.

The order is even. Its half-power is

$$2^{r/2} = 2^6 \equiv 29 \pmod{35}.$$

This is neither 1 nor $-1 \equiv 34$ modulo 35. The final tests give

$$\begin{aligned}\gcd(29 - 1, 35) &= \gcd(28, 35)\\ &= 7,\\ \gcd(29 + 1, 35) &= \gcd(30, 35)\\ &= 5.\end{aligned}$$

Both are proper factors, so this record completes the factorisation. The recovered fraction $1/12$ is not an exactly represented counting phase:

$$\left|\frac{683}{8192} - \frac{1}{12}\right| = \frac{1}{24576}.$$

This is less than $1/(2 \cdot 12^2) = 1/288$, as expected for the continued-fraction criterion. The modular tests certify the order and factors. A sampled, non-dyadic reading does not certify a unique latent numerator s; no such value was assumed in the calculation.

3. Read the physical-qubit label

Write v_j for the column indicating which of g_1, g_2, g_3 acts by X on qubit j. The corresponding Z-type column in g_4, g_5, g_6 is identical. The columns are

$$\begin{aligned} v_1 &= 001, \\ v_2 &= 010, \\ v_3 &= 011, \\ v_4 &= 100, \\ v_5 &= 101, \\ v_6 &= 110, \\ v_7 &= 111. \end{aligned}$$

An X error commutes with the X-type generators and anticommutes with the Z-type generators that contain its qubit. A Z error has the reverse rule. A Y error anticommutes with both types at that qubit. Hence the full six-bit syndromes are

$$\begin{aligned} s(X_j) &= (000, v_j), \\ s(Z_j) &= (v_j, 000), \\ s(Y_j) &= (v_j, v_j). \end{aligned}$$

The nonzero columns are distinct, so the 21 one-qubit Paulis have distinct nonzero syndromes. The observed two halves are equal and nonzero, so the error is Y-type. Since $v_6 = 110$, it is $E = Y_6$. The correction is Y_6 itself.

For every code vector ψ, the relation $g_i\psi = \psi$ implies

$$g_i E\psi = (-1)^{s_i(E)} E\psi.$$

Thus ideal syndrome measurement leaves $E\psi$ unchanged and gives this syndrome with probability one. Since Y_6 is self-adjoint and squares to the identity, correction gives

$$Y_6 E\psi = Y_6^2\psi = \psi.$$

The argument uses no choice of logical basis vector. It preserves every logical superposition.

4. Keep the complement

The code and error-copy projectors are

$$\Pi_0 = \Pi_C = |000\rangle\langle 000| + |111\rangle\langle 111|.$$

$$\Pi_1 = X_2\Pi_C X_2 = |010\rangle\langle 010| + |101\rangle\langle 101|.$$

Their ranges are orthogonal. Since $X_2^\dagger X_2 = \mathbb{1}$ and $\Pi_C X_2 \Pi_C = 0$, the KL condition gives

$$(\alpha_{ij})_{i,j=0}^{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Both eigenvalues are one. The construction therefore uses the isometries $W_0 = \Pi_C$ and $W_1 = X_2\Pi_C$. The complement is

$$\Pi_\perp = \mathbb{1} - \Pi_0 - \Pi_1.$$

It projects onto the span of $|001\rangle$, $|011\rangle$, $|100\rangle$ and $|110\rangle$. In particular, $\Pi_\perp$ is not zero. Take the recovery operators

$$\begin{aligned} R_0 &= \Pi_C, \\ R_1 &= \Pi_C X_2, \\ R_\perp &= \Pi_\perp. \end{aligned}$$

Their completeness is

$$\begin{aligned} R_0^\dagger R_0 + R_1^\dagger R_1 + R_\perp^\dagger R_\perp &= \Pi_0 + \Pi_1 + \Pi_\perp \\ &= \mathbb{1}. \end{aligned}$$

The map $\mathcal{R}(\tau) = \sum_\ell R_\ell \tau R_\ell^\dagger$ is therefore a trace-preserving channel on the full register. Omitting $\Pi_\perp$ would lose all trace on the input $|001\rangle\langle 001|$. The error actions on the code are

	$E_0\Pi_C$	$E_1\Pi_C$
R_0	Π_C	0
R_1	0	Π_C
$R_\perp$	0	0

Linearity now gives $R_0 K_k \Pi_C = a_k \Pi_C$, $R_1 K_k \Pi_C = b_k \Pi_C$ and $\Pi_\perp K_k \Pi_C = 0$. For an arbitrary density operator $\rho = \Pi_C \rho \Pi_C$, this implies

$$(\mathcal{R} \circ \mathcal{N})(\rho) = \sum_k (|a_k|^2 + |b_k|^2)\rho.$$

Compressing the assumed noise completeness to C gives

$$\begin{aligned} \Pi_C \Big(\sum_k K_k^\dagger K_k \Big) \Pi_C &= \sum_k (|a_k|^2 + |b_k|^2) \Pi_C \\ &= \Pi_C. \end{aligned}$$

Since $\Pi_C \neq 0$, the scalar sum is one. Thus the recovery returns ρ, including its logical off-diagonal entries. It does not require the channel to be a classical mixture of the two errors.

For the concrete coherent error, the corrupted vector is

$$V\psi = \frac{\sqrt{3}}{2}\psi - \frac{\mathrm{i}}{2} X_2 \psi.$$

Its two summands occupy orthogonal syndrome subspaces. The probabilities are $3/4$ and $1/4$, independent of α and β. The normalised vectors before correction are ψ and $-\mathrm{i}X_2\psi$. Applying the identity on the first branch and X_2 on the second returns ψ and $-\mathrm{i}\psi$, respectively. Both represent the original logical state. The complement occurs with probability zero for this error, but it is required to define a trace-preserving recovery on other inputs.

5. Multiply before replacing

The generators ZZI and IZZ force all three computational bits to agree, so their common fixed space is spanned by $|000\rangle$ and $|111\rangle$. The generator $-XXX$ fixes their difference. A normalised initial vector is therefore

$$\psi = \frac{|000\rangle - |111\rangle}{\sqrt{2}}.$$

The measured IYI anticommutes with every original generator: its Y meets X, Z and Z on qubit 2. Keep the pivot g_1 temporarily and update the other two rows. Their signs are

$$\begin{aligned} g_2 g_1 &= -(ZX) \otimes (ZX) \otimes X \\ &= -\mathrm{i}^2 YYX \\ &= YYX, \\ g_3 g_1 &= -X \otimes (ZX) \otimes (ZX) \\ &= -\mathrm{i}^2 XYY \\ &= XYY. \end{aligned}$$

Now replace the pivot by $(-1)^b g$. The resulting list is

$$\begin{aligned} &(-1)^b IYI, \\ &YYX, \\ &XYY. \end{aligned}$$

Every pair commutes. Measuring an operator that anticommutes with a stabilizer gives each outcome with probability $1/2$. Writing $\eta = (-1)^b$, the normalised state is

$$\begin{aligned} \psi_b &= \frac{(\mathbb{1} + \eta IYI)\psi}{\sqrt{2}} \\ &= \frac{|000\rangle - |111\rangle + \eta\mathrm{i}\,|010\rangle + \eta\mathrm{i}\,|101\rangle}{2}. \end{aligned}$$

This formula supplies both branches by substituting $b = 0$ or $b = 1$. For a direct check, the operator YYX sends $|000\rangle$ to $-|111\rangle$ and $|111\rangle$ to $-|000\rangle$, and exchanges $|010\rangle$ with $|101\rangle$ without a sign. The same actions on these four basis vectors hold for XYY. Both therefore fix ψ_b. The first updated generator fixes it because $\eta IYI(\mathbb{1} + \eta IYI) = \mathbb{1} + \eta IYI$.

For the follow-up, multiply the first two updated generators:

$$\begin{aligned} (\eta IYI)(YYX) &= \eta YIX, \\ h &= -YIX = -\eta(\eta YIX). \end{aligned}$$

The product ηYIX stabilizes ψ_b. Thus h has eigenvalue $-\eta = (-1)^{b\oplus 1}$ on it. The second measurement is deterministic, with $c = b \oplus 1$. Ignoring either the initial minus sign or the final minus sign of h would change this conclusion.